



ITI5147
Data Exploration and Visualisation

Assessment Title: Project proposal

Project Title: What's the Condition of Education in Indonesia? A Closer Look at Its Quantitative Aspect from Jawa Barat Region

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Project Description and Motivation

5 months ago, on January 10th, 2023, there was an interview with Nadiem Anwar Makarim with Gita Wirjawan¹ that discusses education in Indonesia. One of the topics Gita Wirjawan mentioned is how low Indonesia's PISA score (Programme for International Student Assessment). There are three components in PISA which are reading literacy, mathematics, and science. In 2018, Indonesia only ranked 74th for literacy, 73rd for mathematics, and 71st for science among the other 79 countries that were being measured. (Dian, 2022) From this score alone, it's clear that Indonesia's education quality is not the best.

One important aspect that constitutes education is that every child gets an equal opportunity to pursue education which is measured by the Education Index which is one of the components of the Human Development Index (HDI). HDI is a parameter that's made to assess a country's growth by a tangible approach. By 2010, the method used to calculate HDI is by summing three components which are,

1. Life Expectancy
2. Education or Knowledge Capability
3. Per Capita Income of the Concerned People of a Country.

The second aspect, which is education, is what the Education Index measures. Education Index is measured by two components which are,

- Expected Years of Schooling Index (EYSI) which is the expected years a student is enrolled.
- Mean Years of Schooling Index (MYSI) which is the average number of years a student over the age of 25 has actually received. (Chakravarthi, 2011)

As mentioned above, the education index plays a big role in determining a country's growth. To facilitate children finishing their education, the Indonesian Government released a regulation that mandates every child should get compulsory education for 9 years free of charge, which means from elementary to middle school. For high school, the region is given free rein on whether they

¹ <https://youtu.be/TSER7bEmkXg?t=1654>

should implement it too or not. This regulation is mentioned in “Undang-Undang Republik Indonesia Nomor 20 Tahun 2023 Tentang Sistem Pendidikan”. (Indonesia, 2003) But a recent article in Kompas discusses that compulsory education should be extended to 12 years. Even the Indonesia Ministry of Education is making it a priority for it to be realized. (Ihsan, 2023) Another article from IDN Times stated that the graduation rate for Indonesian students across 4 education levels is as follows.

1. Primary school graduation rate: 100%
2. Middle school graduation rate: 91.9%
3. High school graduation rate: 78%
4. College graduation rate: 19%

It's clear that there's a decrease in every educational stage which could be the reason why the government is pushing for the 12-year compulsory education to be realized. An additional important fact that was discussed in the article is that the Education Index in Indonesia is actually not that low. Of 203 countries across the world, Indonesia ranks 63rd which is at the upper middle of the spectrum. (Zarawaki, 2023)

Of course, there are a lot of other factors that constitute education, one of which is the student-to-teacher ratio. A study conducted in Las Vegas USA concluded that a good student-to-teacher ratio will have a positive impact on students' education. They found that a 15:1 and lower ratio has a positive impact while a 20:1 and higher ratio has a negative impact. (Koc 2015) But, a paper released by students from Universitas Hasanuddin also tackled this problem. They researched factors that affected Education Index in their region in Selayar Islands. From their research, they concluded that while student to teacher ratio has a negative effect on Education Index at the elementary school stage, there's no correlation between Education Index and other educational stages. (Jamal 2021) This goes against another research that founds a positive correlation between the student-to-teacher ratio and with Education Index at the elementary school level. (Mahendra 2016)

To complete the visualization required, datasets were collected from Open Data Jabar². Open Data Jabar is an open website that belongs to Jawa Barat Government that has datasets from regions across Jawa Barat to fulfill the citizen's needs to access accurate and relevant data or information that's readily available. There are 7 data that were collected which is.

1. Indeks Pendidikan Berdasarkan Kabupaten/Kota di Jawa Barat
<https://opendata.jabarprov.go.id/id/dataset/indeks-pendidikan-berdasarkan-kabupatenkota-di-jawa-barat>
2. Jumlah Guru Sekolah Menengah Atas (SMA) Berdasarkan Kabupaten/Kota di Jawa Barat
<https://opendata.jabarprov.go.id/id/dataset/jumlah-guru-sekolah-menengah-atas-sma-berdasarkan-kabupatenkota-di-jawa-barat>

² <https://opendata.jabarprov.go.id/id>

3. Jumlah Murid Sekolah Menengah Atas (SMA) Berdasarkan Kabupaten/Kota di Jawa Barat
<https://opendata.jabarprov.go.id/id/dataset/jumlah-murid-sekolah-menengah-atas-sma-berdasarkan-kabupatenkota-di-jawa-barat>
4. Jumlah Guru Sekolah Menengah Pertama (SMP) Berdasarkan Kabupaten/Kota di Jawa Barat
<https://opendata.jabarprov.go.id/id/dataset/jumlah-guru-sekolah-menengah-pertama-smp-berdasarkan-kabupatenkota-di-jawa-barat>
5. Jumlah Murid Sekolah Menengah Pertama (SMP) Berdasarkan Kabupaten/Kota di Jawa Barat
<https://opendata.jabarprov.go.id/id/dataset/jumlah-murid-sekolah-menengah-pertama-smp-berdasarkan-kabupatenkota-di-jawa-barat>
6. Jumlah Murid Sekolah Dasar (SD) Berdasarkan Kabupaten/Kota di Jawa Barat
<https://opendata.jabarprov.go.id/id/dataset/jumlah-murid-sekolah-dasar-sd-berdasarkan-kabupatenkota-di-jawa-barat>

These six data sources listed the Education Index for each region in the Jawa Barat region according to its regency and city. Also listed are the number of students and teachers there. Each has tabular data, formatted with a CSV file, 8 columns, and around 130-160 rows except for the Education Index which has 322 rows. The data set for elementary school teachers was found but the amount of data is too low so it cannot be used. There might be a better data set around, but it's still not found yet.

Project Value

The three research questions that I aimed to answer with this visualization are.

1. Does the teacher-to-student ratio compared to the Education Index pattern reflect the conclusion derived from the research?
 It's known that the student-to-teacher ratio impacts the quality of education received by the students. But the two research derived in two separate parts in Indonesia have a polarizing conclusion about the effects of the student-to-teacher ratio on the education index. Both of these research also concludes that the student-to-teacher ratio didn't have any effect on the Education Index past elementary school. (Jamal 2021, Mahendra 2016)
 This question is aimed to see if that pattern holds true in the Jawa Barat region.
2. What about the distribution of the teacher-to-student ratio for each Kabupaten and Kota?
 According to the research conducted in Las Vegas, a good student-to-teacher ratio will provide a better-quality education for the students. (Koc 2015) But as it turns out, Indonesia still lacks a lot of teachers. An article at Detik even claims that in 2024 Indonesia will still lack 1.312.759 [Aisyah 2023] So this question is aimed to see how many regions in Jawa Barat have a good teacher-to-student ratio and how many don't have it.
3. What is the pattern of the total of students across the education levels? Does it have a downward trend for all regions in Jawa Barat?
 As mentioned above, the graduation rate of students in Indonesia is decreasing for each

education level. (Zarawaki, 2023) With the push of 12 years of compulsory education, it is hoped that for each education level, the total of students would stay around the same. (Ihsan, 2023) This question is aimed to see if the pattern is true and visualize the region with the total number of students every year.

Data Collection and Wrangling

The data sets gathered are a collection of the Education Index for each region in Jawa Barat from 2015-2020, also the number of students from 2015-2020 and teachers from 2015-2019. Because there were no more recent data sets available, these are the data set that will be used. The data content is as follows.

Data Column Name	Data Description	Which Dataset
Id	Numbering	All of them
Kode Provinsi	The code for the province	All of them
Nama Provinsi	The name of the province	All of them
Kode Kabupaten	The code for the regency	All of them
Nama Kabupaten	The name of the regency it also includes the city region	All of them
Tahun/Tahun Ajaran	Year of the data	All of them
Indeks Pendidikan	Education Index Score	Education Index
Jumlah Murid	Total Students	Students for Elementary, Middle, and High School
Jumlah Guru	Total Teacher	Teacher for Middle and High School
Satuan	Unit of the value	All of them

For the data wrangling, the 2020 data and above will be deleted to accommodate the teacher data since the teacher datasets end in 2019. A new column will also be made which is the student-to-teacher ratio which is gathered by dividing the number of students by the number of teachers respective of the education levels. Finally, all the data will be gathered into one file to make the data visualization easier to do.

Preliminary Data Exploration

From the preliminary data exploration, the mean and median got from the data are.

Name of the Data	Mean	Median
High School Students	25209.27	20161
High School Teachers	1303.311	1145
Middle School Students	66088.43	62039
Middle School Teachers	3135.6	2937
Elementary School Students	166973.5	157450

Sheet 1

YEAR (Tahun Ajaran)

- 2015
- 2016
- 2017
- 2018
- 2019

Nama Kabupaten Kota / Year of Tahun Ajaran

KABUPATEN SUMEDANG

KABUPATEN TASIKMALAYA

KOTA BANDUNG

KOTA BANJAR

KOTA BEKASI

KOTA BOGOR

KOTA CIMAHI

KOTA CIREBON

KOTA DEPOK

Jumlah Girls Sdn

Jumlah Boys Sdn

Bogor and Bandung City, for example, have a spike in the number of high school students and teachers. Both are outliers but true outliers since they represent the number of populations from that area. Because of that these data cannot be deleted/modified.

This research is aimed to see the condition of education in Indonesia based on quantitative measurements. A lot of articles and data support that education in Indonesia is quite bad. This is an attempt to visualize it and spread awareness about it. The main audience target for this visualization is still the government. If the condition is truly as bad as the article and data actually are, this research hoped to help the government see where are the areas that are in need of support, and which are already sufficient enough. This research also hoped to spread more awareness to the public. It is widely known that education is important for the next generation and it's important to visualize how bad or how good it actually is.

Currently, the project state is figuring out what visualization will be used with FDS.

[illegible]

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